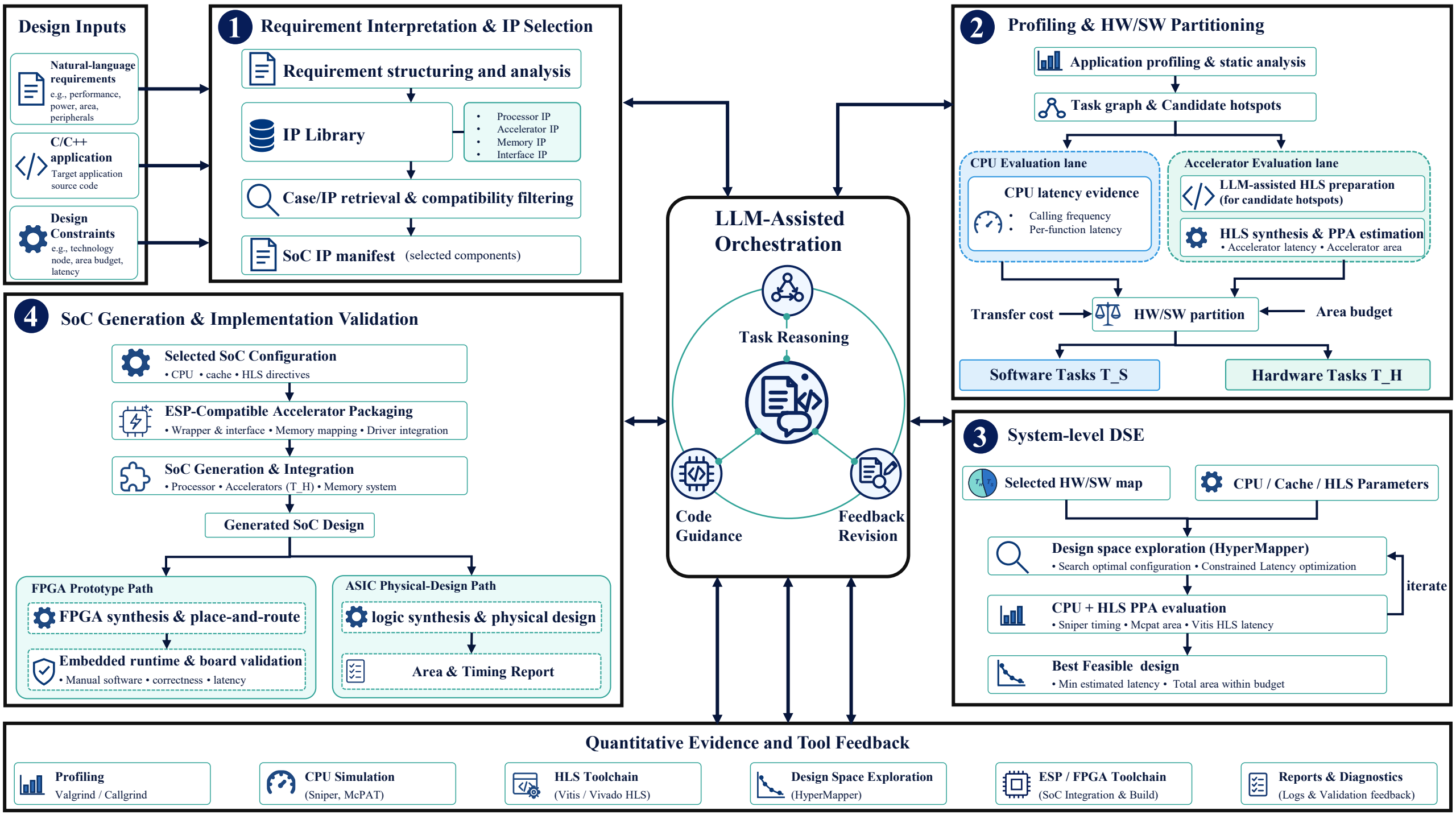

Design Inputs
Natural-language requirements
e.g., performance, power, area, peripherals
C/C++ application
Target application source code
Design Constraints
e.g., technology node, area budget, latency
1 Requirement Interpretation & IP Selection
Requirement structuring and analysis
IP Library
Processor IP
Accelerator IP
Memory IP
Interface IP
Case/IP retrieval & compatibility filtering
SoC IP manifest (selected components)
2 Profiling & HW/SW Partitioning
Application profiling & static analysis
Task graph & Candidate hotspots
CPU Evaluation lane
CPU latency evidence
Calling frequency
Per-function latency
Accelerator Evaluation lane
LLM-assisted HLS preparation (for candidate hotspots)
HLS synthesis & PPA estimation
• Accelerator latency • Accelerator area
Transfer cost
HW/SW partition
Area budget
Software Tasks T_S
Hardware Tasks T_H
LLM-Assisted Orchestration
Task Reasoning
Code Guidance
Feedback Revision
3 System-level DSE
Selected HW/SW map
CPU / Cache / HLS Parameters
Design space exploration (HyperMapper)
• Search optimal configuration • Constrained Latency optimization
CPU + HLS PPA evaluation
• Sniper timing • Mcpat area • Vitis HLS latency
iterate
Best Feasible design
• Min estimated latency • Total area within budget
4 SoC Generation & Implementation Validation
Selected SoC Configuration
• CPU • cache • HLS directives
ESP-Compatible Accelerator Packaging
• Wrapper & interface • Memory mapping • Driver integration
SoC Generation & Integration
• Processor • Accelerators (T_H) • Memory system
Generated SoC Design
FPGA Prototype Path
FPGA synthesis & place-and-route
Embedded runtime & board validation
• Manual software • correctness • latency
ASIC Physical-Design Path
logic synthesis & physical design
Area & Timing Report
Quantitative Evidence and Tool Feedback
Profiling
Valgrind / Callgrind
CPU Simulation
(Sniper, McPAT)
HLS Toolchain
(Vitis / Vivado HLS)
Design Space Exploration
(HyperMapper)
ESP / FPGA Toolchain
(SoC Integration & Build)
Reports & Diagnostics
(Logs & Validation feedback)